AVATAR · early surgery vs watchful waiting · outcome: all-cause death

Asymptomatic severe aortic stenosis · RoB 2 asks up to 22 questions about risk of bias across 5 domains.

Y yes · PY / PN probably yes / no · N no · NI no information

direct support | context | inference (no quote) | not needed (RoB 2 branching)

Sources stored 7 sources stored · 4 cited below

A · Circulation report · 2022 | **B · Eur Heart J report · 2024** | **C · ClinicalTrials.gov · 2015** | **D · Protocol amendment · 2021**

Domain & question	The sources say	AI model answers (interprets evidence)	ARBITER (checks quotes + rules, no AI)	RoB 2 result
Randomisation process Did baseline differences suggest a problem with randomisation? 1.3	**B · p 5** "Table 2 Selected baseline characteristics …" **B · p 5** "Diabetes mellitus 14 (17.9%) 23 (29.1%) .07 …"	**No** "The age difference is 1 year in medians. This is compatible with chance …"	1.1 Y · 1.2 Y · 1.3 N	**Low**
Deviations from intended interventions Did the trial itself cause departures from the intended treatment? 2.3	**B · p 6** "Symptom onset … AS progression …" **D · p 2** "delayed surgery … due to COVID -19 …" **A · p 4** "referred … AS-related symptoms …" **B · p 9** "physical visits … phone follow-up. …"	**Probably no** "Crossover from conservative care to surgery (35/79) was driven by protocol-defined indications …"	2.1 Y · 2.2 Y · 2.3 PN · 2.4 · 2.5 · 2.6 Y · 2.7	**Low**
Missing outcome data Were outcome data available for nearly all participants? 3.1	**B · p 6** "All patients were included in the analysis. … One patient … lost to follow-up … censored …"	**Yes** "1/157 (0.6%) has incomplete outcome data."	3.1 Y · 3.2 · 3.3 · 3.4	**Low**
Outcome measurement Did assessors know the treatment? Could that affect how deaths were recorded? 4.3, 4.4	**B · p 3** "members were not blinded to … the treatment allocation during events review. …"	**Yes** **Probably no** *"All-cause death … involves no judgement"*	4.1 PN · 4.2 PN · 4.3 Y · 4.4 PN · 4.5	**Low**
Selection of the reported result Was the analysis plan fixed before the results were known? 5.1	**C · p 1** "all cause death, … (MACE) including: …" **A · p 4** "Prespecified secondary … Time to death …" **B · p 5** "all analyses … intention-to-treat principle, …" **D · p 1** "-Amendment to surgical delays- …"	**No information** "published protocol … is not among the captured Sources … the report statements are not date-stamped before unblinded data."	5.1 NI · 5.2 PN · 5.3 PN decided by question 5.1	**Some concerns**

? Open point: Reasons for declining surgery are not described in detail.

Weighed against: Diabetes was less common in early surgery (17.9% vs 29.1%), … but P=.07 is compatible with chance

Weighed against: pandemic-related follow-up substitution by phone and lengthened AVR delays could have increased clinical risk. … not the trial context … no deaths occurred while awaiting scheduled surgery

? Open point: Whether the published protocol listed time to all-cause death and its Cox analysis, and its date relative to unblinded data.

Overall Any domain with Some concerns → overall Some concerns (no domain High) — **Some concerns**

This trial: 15 questions answered · 7 not needed · 16 quotes, all found in the stored sources

All of this appears in ARBITER's HTML report for the reviewer and is saved in one audit file (.arbiter.zip), checked by an independent verifier.

Published review: Low · Both expert adjudicators: Some concerns